Unleash your potential

Module
Presenter's Manual
for

Introduction to Dart Programming

Effective from: June 2024
Ver. 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

Table of Contents

1. Introduction

At the end of this course, students will be able to:

- Explain Dart Programming and its installation.
- Understand Data types in Dart.
- Explain Control flow statements in Dart programming.
- Describe OOP in Dart.
- Applying OOP Concepts in Dart.
- Discuss on Collections and Exceptions in Dart.
- Explain the use of Synchronous and Asynchronous Programming in Dart.
- Explain Dart Package, its necessity, types, and creation of user-defined package in Dart.
- Understanding the structure of JSON document in Dart.

2. Information on Session Allocation

Module	Online (No. of Hrs)
Introduction to Dart Programming	**16**

Throughout this Presenter's Manual, the module **Introduction to Dart Programming** will be referred to as **IDP.**

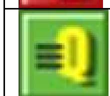
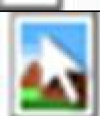
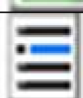
3. Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	***Download Book -*** Student has the option to download the subject related e-book and read offline.
	***Glossary -*** Student can access a list of subject related specialized words with their definitions.
	***FAQ -*** Student can access frequently asked questions and their answers.
	***Practice 4 Me -*** Student can test and evaluate their understanding of module related topics.
	***Work Assignments -*** Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	***References -*** Student can access additional subject related material for reading.
	***Feedback -*** Student can provide feedback on the course material.
	***Ask to Learn –*** Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- **A Session is of 2 hours duration**

➢ Week-Wise Schedule

Wk	Day 1	Day 2	Day 3	Day 4
1	**Session 1** **IDP– TL1**	**Session 2** **IDP – TL2**	**Session 3** **IDP – TL3**	**Session 4** **IDP – TL4**
2	**Session 5** **IDP – TL5**	**Session 6** **IDP – TL6**	**Session 7** **IDP – TL7**	**Session 8** **IDP – TL8**

TL: Online Session

5. Session Coverage

Sess No.	Session Title	Session Details	Deliverables' Mapping
1	**IDP– TL1**	All the topics as listed below from Session 1 and Session 2 of *Introduction to Dart Programming* book should be covered in this session. **Session 1 – Introduction to Dart** ➢ Define Dart and explain the history of Dart ➢ Describe the features of Dart ➢ List advantages of Dart ➢ Explain the installation of Dart SDK in Android Studio ➢ Create a simple Dart application ➢ Explain identifiers in Dart ➢ List the basic keywords in Dart **Session 2 – Variables and Data Types** ➢ Understanding Data types in Dart ➢ Explain Variables, String, Boolean, Map, Dynamic, Constants in Dart	**Introduction to Dart Programming** SG - Session 1 & 2 XP – Session 1 & 2 TG- Session 1 & 2
2	**IDP– TL2**	The **Try It Yourself** questions of Session 1 and Session 2 of *Introduction to Dart Programming* book should be covered in this session.	**Introduction to Dart Programming** Session 1 & 2
3	**IDP– TL3**	All the topics as listed below from Session 3 and Session 4 of *Introduction to Dart Programming book* should be covered in this session. **Session 3 – Operators amd Control Flow Statements** ➢ Define Operators ➢ Explain Control Flow	**Introduction to Dart Programming** SG - Session 3 & 4 XP – Session 3 & 4 TG - Session 3 & 4

<table>
<tr><th>Sess No.</th><th>Session Title</th><th>Session Details</th><th>Deliverables' Mapping</th></tr>
<tr><td></td><td></td><td>Statements in Dart
➢ Define Decision-making Statement in Dart
➢ Define Looping Statement in Dart
➢ Identify Jump Statement in Dart

<u>Session 4 – Object-Oriented Programming in Dart</u>

➢ Explain OOP in Dart
➢ Explain Methods in Dart
➢ Define class, objects, inheritance, polymorphism, interface, and abstraction
➢ Describe Getter and Setter Methods in Dart
➢ Applying OOP Concepts in Dart
➢ Explain Constructors in Dart</td><td></td></tr>
<tr><td>4</td><td>IDP– TL4</td><td>The Try It Yourself questions of Session 3 and Session 4 of Introduction to Dart Programming book should be covered in this session.</td><td>Introduction to Dart Programming

Session 3 & 4</td></tr>
<tr><td>5</td><td>IDP– TL5</td><td>All the topics as listed below from Session 5 and Session 6 of Introduction to Dart Programming book should be covered in this session.

<u>Session 5 – Collections and Exceptions in Dart</u>
➢ Define Collection and its usage.
➢ Explain Types of Collections in Dart
➢ Define and Outline Exception in Dart
➢ Explain Exception handling in Dart
➢ Elaborate on Errors in Dart

<u>Session 6 – Synchronous and Asynchronous Programming in</u></td><td><u>Introduction to Dart Programming</u>

SG - Session 5 & 6
XP – Session 5 & 6
TG - Session 5 & 6</td></tr>
</table>

Sess No.	Session Title	Session Details	Deliverables' Mapping
		Dart ➢ Explain Synchronous and Asynchronous Programming in Dart ➢ Learn asynchronous programming with future keyword ➢ Learn asynchronous programming with async and await keyword	
6	**IDP– TL6**	The **Try It Yourself** questions of Session 5 and Session 6 of *Introduction to Dart Programming* book should be covered in this session.	**Introduction to Dart Programming** Session 5 & 6
7	**IDP– TL7**	All the topics as listed below from Session 7 and Session 8 of *Introduction to Dart Programming book* should be covered in this session. **Session 7 –Packages in Dart** ➢ Define Dart Package and its usage ➢ Explain the Types of Dart Packages **Session 8 – JSON in Dart** ➢ Identify JSON in Dart ➢ Explain the structure of a JSON document ➢ Describe Encoding and Decoding of JSON in Dart	**Introduction to Dart Programming** SG - Session 7 & 8 XP – Session 7 & 8 TG - Session 7 & 8
8	**IDP– TL8**	The **Try It Yourself** questions of Session 7 and Session 8 of *Introduction to Dart Programming* book should be covered in this session.	**Introduction to Dart Programming** Session 7 & 8

6. Library References

➢ **Dart for Absolute Beginners** by David Kopec
➢ **The Dart Programming Language** by Gilad Bracha
➢ **Learning Dart** by Ivo Balbaert and Dzenan Ridjanovic

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